Nonequilibrium free energies of a pulled polymer: Jarzynski, Crooks, and the cost of chain length

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Abstract

The Jarzynski equality and the Crooks fluctuation theorem relate the work performed in arbitrarily fast, irreversible processes to equilibrium free-energy differences. They are exact, but the exponential averages they involve are dominated by rare trajectories, which limits their practical use. We study this limitation in a minimal model where every quantity can be checked against an exact answer: a two-dimensional bead–spring chain, anchored at one end and stretched by a harmonic trap moving through a Langevin heat bath. Exact equilibrium sampling and a numerically exact free-energy reference allow us to separate the physics from the statistics. With 10^5 pulls per condition we find that the dissipated work, measured in units of $k_{\mathrm{B}}T$, is controlled by a single combination $\beta W_{\mathrm{diss}} \simeq (\pi^2/6)\, N/(\tau/\tau_1)$ of the chain length N and the pulling time τ relative to the slowest relaxation time τ_1. At fixed reduced speed it is independent of temperature, and the work distributions at different temperatures collapse. It grows linearly with N, so the number of pulls needed by the Jarzynski estimator grows exponentially with chain length. At $N = 40$ the one-sided estimate is off by $1.5\, k_{\mathrm{B}}T$ even with 10^5 pulls, while Bennett's acceptance ratio, which combines forward and reverse pulls, recovers the exact free energy to within $0.01\, k_{\mathrm{B}}T$. Plotting the distribution of the Jarzynski weight $e^{-\beta W}$ shows directly why: the exponential average is carried by trajectories that are typical of the *reverse* process and are essentially never sampled in the forward one.

1 Introduction

Free-energy differences determine equilibrium constants, binding affinities and the mechanical stability of molecules. Classically they are measured along reversible, quasi-static paths, on which the work equals the free-energy change. Jarzynski showed in 1997 that this restriction is unnecessary: for a system that starts in equilibrium and is driven by an external parameter along *any* protocol,

$$\left\langle e^{-\beta W} \right\rangle = e^{-\beta \Delta F}, \tag{1}$$

where W is the work done in one realisation, $\beta = 1/k_{\mathrm{B}}T$, and the average runs over repetitions of the process [1]. Crooks later derived the stronger statement that forward and time-reversed work distributions are related by $P_F(W)/P_R(-W) = e^{\beta(W-\Delta F)}$ [2]. Both relations have been verified in single-molecule pulling experiments on RNA hairpins [3, 4] and are now standard tools in stochastic thermodynamics [5].

Their practical limitation is statistical. The average in (1) is dominated by rare realisations in which the work falls *below* ΔF. Such realisations are exponentially rare in the dissipated work $W_{\mathrm{diss}} = \langle W \rangle - \Delta F$, so the number of repetitions needed grows like $e^{\beta W_{\mathrm{diss}}}$ [6, 7]. Since dissipation is generally extensive, this suggests an exponential cost in system size. That cost is the usual reason given for why these relations are tools for single molecules rather than for macroscopic systems.

In this work we make these statements quantitative in a model simple enough to be solved exactly at equilibrium, yet rich enough to have non-trivial, entropic elasticity: a two-dimensional chain of N beads, anchored at one end and stretched by a moving optical-tweezer-like trap. The model has three properties that are rarely available together. Its equilibrium free energy can be computed exactly for any N and T. Its initial equilibrium ensemble can be sampled exactly, without equilibration runs. And a Gaussian limit of the model gives the full work distribution of the discretised dynamics in closed form. Every estimate below can therefore be compared with the truth, and simulation errors can be separated from estimator errors.

Our main findings are the following. (i) A simple affine-stretching argument predicts the dissipated work to within about 12% for $N \geq 10$. In units of $k_\mathrm{B}T$ it depends only on N and on the pulling time in units of the chain's relaxation time. (ii) As a consequence, the work distributions of an entropic chain at different temperatures collapse onto one curve when the pulling time is scaled with the relaxation time. (iii) The one-sided Jarzynski estimator fails exponentially in chain length, and its bootstrap error bars are over-confident by a factor of about three. (iv) The Crooks theorem holds with log-ratio slopes within 2% of the predicted value. The two-sided Bennett estimator recovers ΔF to $0.01\,k_\mathrm{B}T$ precisely where the one-sided estimators fail. (v) Visualising the distribution of $e^{-\beta W}$ explains (iii) and (iv) directly through the Crooks identity.

2 Model and methods

Derivations are given in a companion theory document [11]; here we state what is needed to follow the results.

2.1 Anchored chain in a moving trap

The chain consists of beads $\boldsymbol{r}_0, \ldots, \boldsymbol{r}_N \in \mathbb{R}^2$ of mass m. Bead 0 is fixed at the origin, and bead N is held by a harmonic trap of stiffness k_t centred at $\boldsymbol{\lambda}$:

$$H(\boldsymbol{x}; \boldsymbol{\lambda}) = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m} + \sum_{i=1}^{N} \frac{k}{2}\big(|\boldsymbol{r}_i - \boldsymbol{r}_{i-1}| - b\big)^2 + \frac{k_t}{2}|\boldsymbol{r}_N - \boldsymbol{\lambda}|^2. \tag{2}$$

We consider two versions. For $b = 0$ (Gaussian or Rouse chain) all forces are linear and the elasticity is energetic. For $b = 1$ with stiff bonds ($k = 100$, so bond-length fluctuations are $\sim 10\%$ at $k_\mathrm{B}T = 1$) the chain is nearly freely jointed and its elasticity is entropic. The chain is a phantom chain: non-bonded beads do not interact. The trap moves at constant speed along x, $\boldsymbol{\lambda}(t) = (x_0 + (x_1 - x_0)t/\tau, 0)$, from $x_0 = 0.2L$ to $x_1 = 0.7L$, where $L = Nb$ is the contour length ($L = 0.3N$ for $b = 0$). Units are $m = b = \varepsilon = 1$; we use $k_t = 10$ and friction rate $\gamma = 1$ throughout.

2.2 Dynamics, work and fluctuation theorems

Each mobile bead follows underdamped Langevin dynamics, $\mathrm{d}\boldsymbol{p}_i = -\nabla_i H\,\mathrm{d}t - \gamma \boldsymbol{p}_i\,\mathrm{d}t + \sqrt{2\gamma m k_\mathrm{B}T}\,\mathrm{d}\boldsymbol{B}_i$. The friction and noise model collisions with the solvent; their strengths are tied by the fluctuation–dissipation relation, which makes the canonical distribution $\rho_\mathrm{eq} \propto e^{-\beta H}$ stationary at every fixed $\boldsymbol{\lambda}$. The work done by the moving trap in one pull is

$$W = \int_0^\tau \frac{\partial H}{\partial \boldsymbol{\lambda}} \cdot \dot{\boldsymbol{\lambda}}\,\mathrm{d}t = -k_t \int_0^\tau (\boldsymbol{r}_N - \boldsymbol{\lambda}) \cdot \dot{\boldsymbol{\lambda}}\,\mathrm{d}t. \tag{3}$$

Equation (1) holds whenever the chain starts in equilibrium and the dynamics at frozen $\boldsymbol{\lambda}$ preserves the canonical distribution. The Crooks theorem additionally involves the reverse

	$N=5$	$N=10$	$N=20$	$N=40$	$k_\mathrm{B}T=0.25$	$k_\mathrm{B}T=2$
exact $\beta\Delta F$	1.98	4.42	9.49	19.69	4.62	4.18
entropic spring	2.10	4.29	8.65	17.39	4.44	4.09

Table 1: Free-energy differences of the stiff-bond chain for the protocol $0.2L \to 0.7L$ (left: $k_\mathrm{B}T = 1$; right: $N = 10$), in units of $k_\mathrm{B}T$.

protocol ($x_1 \to x_0$ in the same time, starting in equilibrium at x_1). It implies

$$\frac{e^{-\beta W} P_F(W)}{\langle e^{-\beta W}\rangle_F} = P_R(-W), \tag{4}$$

which we use repeatedly. The share of the exponential average contributed by forward works near W equals the mirrored reverse work distribution.

We compare three estimators of ΔF:

- the forward Jarzynski estimator $\widehat{\Delta F}_J = -k_\mathrm{B}T \ln\left(M^{-1}\sum_m e^{-\beta W_m}\right)$;
- its reverse counterpart $+k_\mathrm{B}T \ln\left(M^{-1}\sum_m e^{-\beta W^R_m}\right)$;
- Bennett's acceptance ratio (BAR), the root of $\sum_F \mathrm{f}(\beta(W_m - \Delta F)) = \sum_R \mathrm{f}(\beta(W^R_m + \Delta F))$ with $\mathrm{f}(x) = 1/(1+e^x)$ for equal sample sizes. BAR is the minimum-variance and maximum-likelihood combination of both directions [8, 9].

As diagnostics we report the effective sample size $M_\mathrm{eff} = (\sum w_m)^2/\sum w_m^2$ of the weights $w_m = e^{-\beta W_m}$ and the convergence of $\widehat{\Delta F}_J$ with the number of pulls used.

2.3 Exact reference free energies

In bond variables $\boldsymbol{u}_i = \boldsymbol{r}_i - \boldsymbol{r}_{i-1}$ the bonds are independent except for the trap, so the partition function is a convolution, $Z(\boldsymbol{\lambda}) \propto (g^{*N} * w)(\boldsymbol{\lambda})$, of the bond Boltzmann factor g with itself N times and with the trap factor w. For $b = 0$ this gives $\Delta F = \frac{1}{2}k_\mathrm{eff}(x_1^2 - x_0^2)$ with $1/k_\mathrm{eff} = N/k + 1/k_t$, independent of temperature. For $b = 1$ we evaluate the radial two-dimensional convolutions numerically in log space. The equivalent Fourier-space formula loses all precision through cancellation once $\beta\Delta F \gtrsim 10$. The log-space method reproduces the closed form to 10^{-4}. Table 1 lists the values used. For the entropic chain $\Delta F/k_\mathrm{B}T$ is nearly temperature independent, as expected for rubber elasticity. The Gaussian "entropic-spring" approximation misses up to 12% (finite extensibility at $N = 40$) and even has the wrong sign of error for $N = 5$, where a short two-dimensional chain is strongly non-Gaussian.

2.4 Simulation

Integrator. The Langevin equations are integrated with the BAOAB splitting [10], $\Delta t = 0.01$. The trap position is updated in discrete steps, and the work is accumulated as the energy jump $H(\boldsymbol{x};\boldsymbol{\lambda}_{k+1}) - H(\boldsymbol{x};\boldsymbol{\lambda}_k)$ at fixed configuration before each integrator step. This switch-then-relax construction satisfies a discrete-time version of (1), up to the $O(\Delta t^2)$ error with which a BAOAB step preserves the canonical distribution. For the Gaussian chain, propagating means and covariances through the linear update rules gives the exact work statistics of the discrete algorithm. The residual violation of (1) is $8 \times 10^{-4}\, k_\mathrm{B}T$ and scales as Δt^2. Halving Δt for the stiff chain changes no estimate beyond its statistical scatter (Appendix A).

Exact initial states. Pulls must start in equilibrium. Instead of equilibrating each chain dynamically, we draw exact samples by rejection. Chains are proposed with independent bonds that feel a constant pulling force f, and each is accepted with probability $e^{-\beta k_t |\boldsymbol{R}-\boldsymbol{\lambda}'|^2/2}$, where

$\boldsymbol{R}$ is the end-to-end vector and $\boldsymbol{\lambda}' = \boldsymbol{\lambda} - (f/k_t)\hat{\boldsymbol{x}}$. The procedure is exact for any f. Choosing f equal to the chain tension at the target extension raises the acceptance rate from $\sim 10^{-9}$ to $\sim 1\%$ for the most stretched states ($N = 40$ at $0.7L$).

Relaxation time and protocols. Pulling times are measured in units of $\tau_1 = \zeta/\lambda_{\min}(K)$, the slowest relaxation time of the linearised (Rouse) chain. Here $\zeta = m\gamma$ and K is the stiffness matrix of a chain of entropic springs $\kappa = 2k_{\mathrm{B}}T/\langle u^2\rangle$ with the stiff trap at its end. Because the trap nearly clamps the end, $\tau_1 \simeq \zeta N^2 \langle u^2\rangle /2\pi^2 k_{\mathrm{B}}T$, giving $\tau_1 = 1.45, 5.46, 21.3, 84.3$ for $N = 5, 10, 20, 40$ at $k_{\mathrm{B}}T = 1$. In a slow pull the chain stretches affinely, with bead i moving at speed $v\, i/N$. The friction power is then $\zeta N v^2/3$, which yields the estimate

$$\beta W_{\mathrm{diss}} \simeq \frac{\zeta N (x_1 - x_0)^2}{3k_{\mathrm{B}}T\tau} = \frac{\pi^2}{6}\,\frac{N}{\tau/\tau_1}. \tag{5}$$

Statistics. Each condition uses $M = 10^5$ independent pulls, integrated in parallel on a GPU. All code is covered by unit tests, which check forces, exact sampling, the exact Gaussian-chain moments and Crooks relations, the first law per trajectory, and the estimators on synthetic data.

3 Results

3.1 Validation on the Gaussian chain

For the Gaussian chain the simulated mean and variance of W agree with the exact values of the discrete algorithm to 0.1% and 1%, and the whole distributions coincide (Fig. 1a). With 10^5 pulls, the Jarzynski estimator gives 10.134 ± 0.015 for the slow pull and 10.19 ± 0.14 for the fast one ($\Delta F = 10.125$), although the mean work exceeds ΔF by 1.7 and $5.3\, k_{\mathrm{B}}T$. The convergence curves (Fig. 1b) show the characteristic behaviour: with few pulls the estimate sits near $\langle W\rangle$, and it approaches ΔF only once rare low-work pulls have been sampled. That takes about 100 times more pulls for the faster protocol.

The lower panels show why. The weight $x = e^{-\beta(W-\Delta F)}$ has mean exactly one, but for Gaussian work the pulls and the contributions to the mean are displaced from each other by $\beta^2\sigma_W^2$ in $\ln x$. That is 1.5 decades for the slow pull, where the sampled contributions reproduce the exact ones, and 4.6 decades for the fast pull. In the fast case the true mean is dominated by values of x around $10^{2.3}$, reached by only 0.06% of the pulls. The sampled contributions reduce to a handful of spikes.

3.2 Pulling speed: dissipation and where the average comes from

For the entropic chain, the dissipated work at $\tau/\tau_1 = 1, 3, 30$ is $\beta W_{\mathrm{diss}} = 18.3$, 6.2 and 0.61. The affine estimate (5) gives 16.4, 5.5 and 0.55: it is low by a constant 11–12%, so the $1/\tau$ scaling holds down to $\tau = \tau_1$. The near-equilibrium relation $\sigma_W^2 = 2k_{\mathrm{B}}TW_{\mathrm{diss}}$ holds to 3% for the slowest pull but is exceeded by factors 1.2 and 1.5 for the faster ones, where the distribution also acquires a positive skew. As a result the second-cumulant estimate $\langle W\rangle - \beta\sigma_W^2/2$, exact for Gaussian work, fails by $10\, k_{\mathrm{B}}T$ at $\tau = \tau_1$.

The Jarzynski estimator converges for $\tau = 30\tau_1$ (4.415 ± 0.005) and, with 10^5 pulls, for $3\tau_1$ (4.55 ± 0.10), but not for $\tau = \tau_1$. Panels (c–e) make the reason visible, using the reverse pulls to obtain the *exact* distribution of contributions to $\langle e^{-\beta W}\rangle$ without any rare-event sampling. At $\tau = \tau_1$ half of the true average comes from weights $x > 10^{5.9}$, while the largest weight among all 10^5 forward pulls is $10^{4.3}$: no forward pull reaches the region that matters. The sample mean is 0.57 of the true mean and is carried by three pulls (33%, 27% and 14% of it). The resulting estimate, 4.99, still looks reasonable, because the logarithm compresses the error: missing half of the average costs only $k_{\mathrm{B}}T \ln 2$. Nothing in the forward data alone signals the problem.

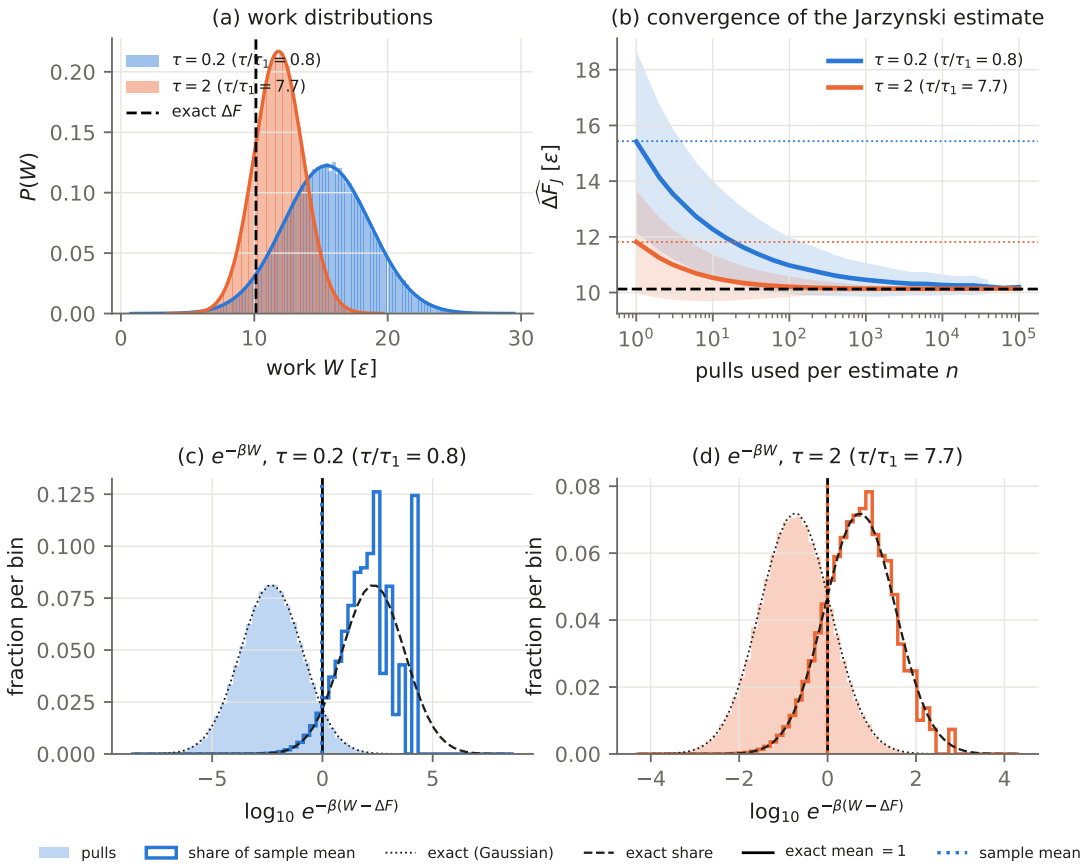


Figure 1: Gaussian chain ($b = 0$, $N = 10$, $k_\mathrm{B}T = 1$, $\Delta F = 10.125$). (a) Work distributions for a fast ($\tau = 0.8\tau_1$) and a slow ($\tau = 7.7\tau_1$) pull; lines are the exact distributions of the discrete algorithm. (b) Jarzynski estimate using n pulls (mean and spread over disjoint blocks); dotted lines mark $\langle W \rangle$. (c, d) Distribution of the normalised Jarzynski weight $x = e^{-\beta(W-\Delta F)}$ (exact mean 1): filled, fraction of pulls; solid, share of the sample mean; dashed, exact share.

3.3 Temperature: entropic elasticity and data collapse

The two chains respond to temperature in opposite ways (Fig. 3b). For the Gaussian chain ΔF is temperature independent, as is the mean work, because the mean response of a linear system is deterministic. The bath only broadens the distribution, $\sigma_W^2 \propto T$, so $\beta W_\mathrm{diss} \propto 1/T$. For the entropic chain $\Delta F \propto T$: a hotter chain is harder to stretch.

When the pulling time is scaled with the temperature-dependent relaxation time, $\tau = 4\tau_1(T)$, the distributions of βW at four temperatures spanning a factor of eight fall on top of each other (Fig. 3a), and βW_diss stays between 4.2 and 4.9. This follows from Eq. (5). Both ΔF and W_diss are proportional to T at fixed τ/τ_1, so temperature merely changes the energy unit. The residual drift comes from the slight thermal stretching of the bonds. At a fixed pulling *time*, by contrast, the dissipated work is set by friction and speed and is nearly temperature independent, so $\beta W_\mathrm{diss} \propto 1/T$ (18.4, 9.3, 4.6, 2.2). At $k_\mathrm{B}T = 0.25$ it exceeds $\ln M$ and the estimate is 20% too high. Experiments in a cold bath are harder only because the same pulling speed is relatively faster.

3.4 Chain length: an exponential cost

At fixed relative extension both ΔF and $\langle W \rangle$ are nearly extensive in N, while σ_W grows like $\sqrt{N}$: relative work fluctuations shrink as $1/\sqrt{N}$, as expected from the macroscopic limit. At fixed reduced speed the dissipated work grows linearly, $\beta W_\mathrm{diss} = 3.1$, 4.6, 8.3 and 16.0 for

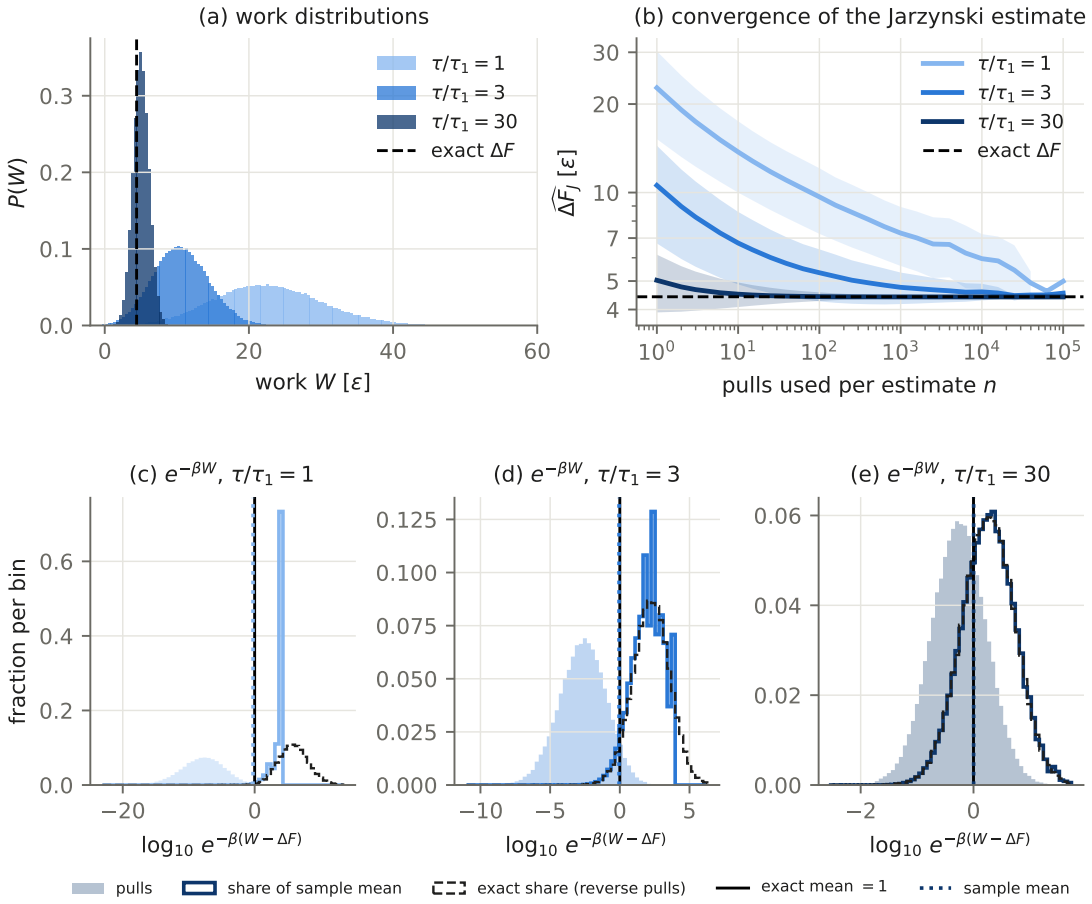


Figure 2: Stiff-bond chain ($b = 1$, $N = 10$, $k_\mathrm{B}T = 1$, $\Delta F = 4.420$, $\tau_1 = 5.46$) pulled at three speeds. (a) Work distributions. (b) Convergence of the Jarzynski estimate. (c–e) Distribution of $x = e^{-\beta(W-\Delta F)}$ as in Fig. 1; the exact share of the mean is obtained from 10^5 reverse pulls through Eq. (4).

$N = 5$, 10, 20 and 40, and approaches the affine estimate $\pi^2 N/24$ (within 1% at $N = 20$, 3% at $N = 40$). The number of pulls needed therefore grows exponentially with chain length. The Jarzynski estimate is accurate to $0.03\,k_\mathrm{B}T$ up to $N = 10$, biased by $+0.38\,k_\mathrm{B}T$ at $N = 20$ and by $+1.47\,k_\mathrm{B}T$ at $N = 40$, where $M_\mathrm{eff} = 3.6$. At fixed computational budget the bias grows steeply with N (Fig. 4c). Keeping βW_diss fixed would require $\tau \propto N\tau_1 \propto N^3$.

The lower row shows the mechanism. As N grows, the pulls move to lower weights and the exact contributions to the mean move to higher ones. At $N = 40$, 81% of the true average lies beyond the largest weight observed in 10^5 forward pulls, and a single pull carries half of the sample mean. In the failing cases the bootstrap error bars do not cover the exact value (3.4σ off at $N = 20$, 2.7σ at $N = 40$). Where the true spread of the estimator could be measured, it was 2–3 times larger than the bootstrap estimate, because resampling the observed pulls cannot account for regions that were never sampled (Appendix A).

3.5 Crooks theorem and two-sided estimation

For every speed the forward and mirrored reverse distributions intersect at ΔF (Fig. 5a–c), and the logarithm of their ratio is a straight line of slope 0.98–0.99 (prediction: 1), the same for all speeds (Fig. 5d). This holds even though the distributions change completely between $\tau = \tau_1$ and $30\tau_1$.

Forward and reverse pulls are not symmetric. The reverse pull dissipates less ($\beta W^R_\mathrm{diss} = 13.6$ against $\beta W^F_\mathrm{diss} = 18.3$ at $\tau = \tau_1$), its distribution is narrower ($\sigma_R = 5.2$ against $\sigma_F = 7.5$),

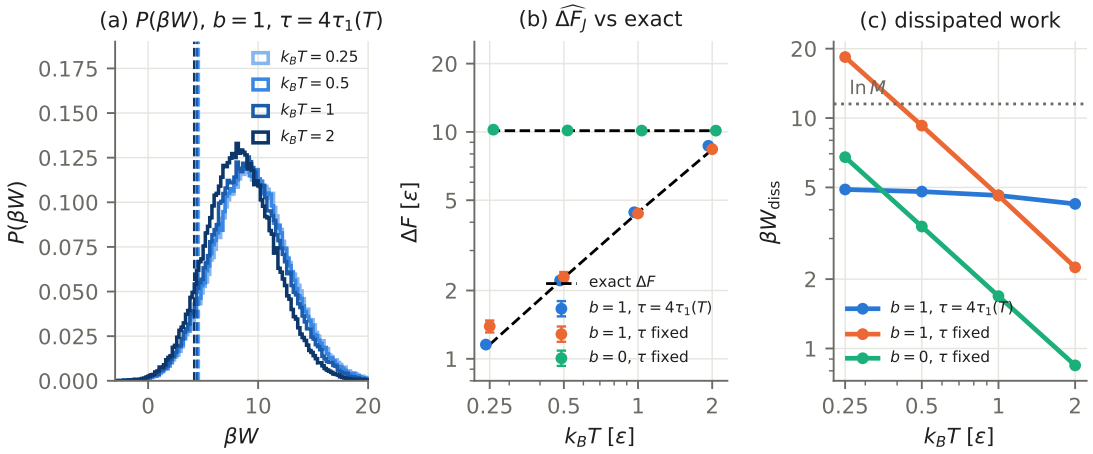


Figure 3: Temperature dependence ($N = 10$). (a) Distributions of βW for the stiff-bond chain at fixed reduced speed $\tau = 4\tau_1(T)$; dashed lines mark $\beta\Delta F$. (b) Jarzynski estimates against the exact ΔF (dashed) for the stiff-bond chain at fixed τ/τ_1 and at fixed τ, and for the Gaussian chain. (c) Dissipated work in units of $k_{\mathrm{B}}T$; dotted line: $\ln M$.

and it obeys the linear-response relation $\sigma_W^2 = 2k_{\mathrm{B}}TW_{\mathrm{diss}}$ within 10%, whereas the forward pull exceeds it by 54%. Both distributions remain close to Gaussian, with skewness below 0.22. The asymmetry disappears near equilibrium, where Crooks forces $W^F_{\mathrm{diss}} = W^R_{\mathrm{diss}}$. A plausible interpretation is that a fast forward pull drives the end of the chain into the stiff, finite-extensibility regime, where the force is very sensitive to the chain's conformation, whereas a reverse pull releases tension towards the soft, Gaussian regime.

The asymmetry matters for estimation. By Eq. (4), the forward estimator needs $M \gtrsim e^{\beta W^R_{\mathrm{diss}}}$ pulls to sample the region that dominates its average, and the reverse estimator needs $e^{\beta W^F_{\mathrm{diss}}}$. At $N = 40$ both conditions fail with 10^5 pulls. The one-sided estimates are off by $+1.47$ (forward) and $-0.76\,k_{\mathrm{B}}T$ (reverse), in opposite directions as expected (Fig. 5e). BAR, which only needs the two distributions to overlap near ΔF, gives 19.688 ± 0.039 against the exact 19.694. The overlap region contains only 0.4% of the forward and 0.3% of the reverse pulls, a few hundred each, which suffices. BAR is within $0.01\,k_{\mathrm{B}}T$ of the exact value for all chain lengths. In the most demanding single case ($N = 10$, $\tau = \tau_1$) it is $0.08\,k_{\mathrm{B}}T$ low, compared with $+0.57$ and $-0.54\,k_{\mathrm{B}}T$ for the one-sided estimates.

4 Discussion

Three points emerge that go beyond verifying the fluctuation theorems.

First, a single dimensionless number controls the practical usefulness of the Jarzynski equality in this system: $\beta W_{\mathrm{diss}} \simeq (\pi^2/6)N/(\tau/\tau_1)$. Temperature enters only through the relaxation time, and chain length enters linearly. The exponential cost in system size, usually stated qualitatively, here takes an explicit form. At the reduced speed used here ($\tau = 4\tau_1$) the required number of pulls grows like $e^{0.4N}$. Doubling the chain from 20 to 40 beads turned a slightly biased estimate into a failed one.

Second, the standard diagnostics are unreliable exactly where they are needed. The second-cumulant estimate fails as soon as the work distribution is skewed. The bootstrap underestimates the error because it cannot see unsampled regions. And the Jarzynski estimate itself is deceptively stable, because the logarithm compresses large relative errors in the exponential average. The distribution of $e^{-\beta W}$, combined with reverse pulls through the Crooks identity, provides a direct check: if the exact contributions to the average lie beyond the largest sampled weight, the forward estimate cannot be trusted.

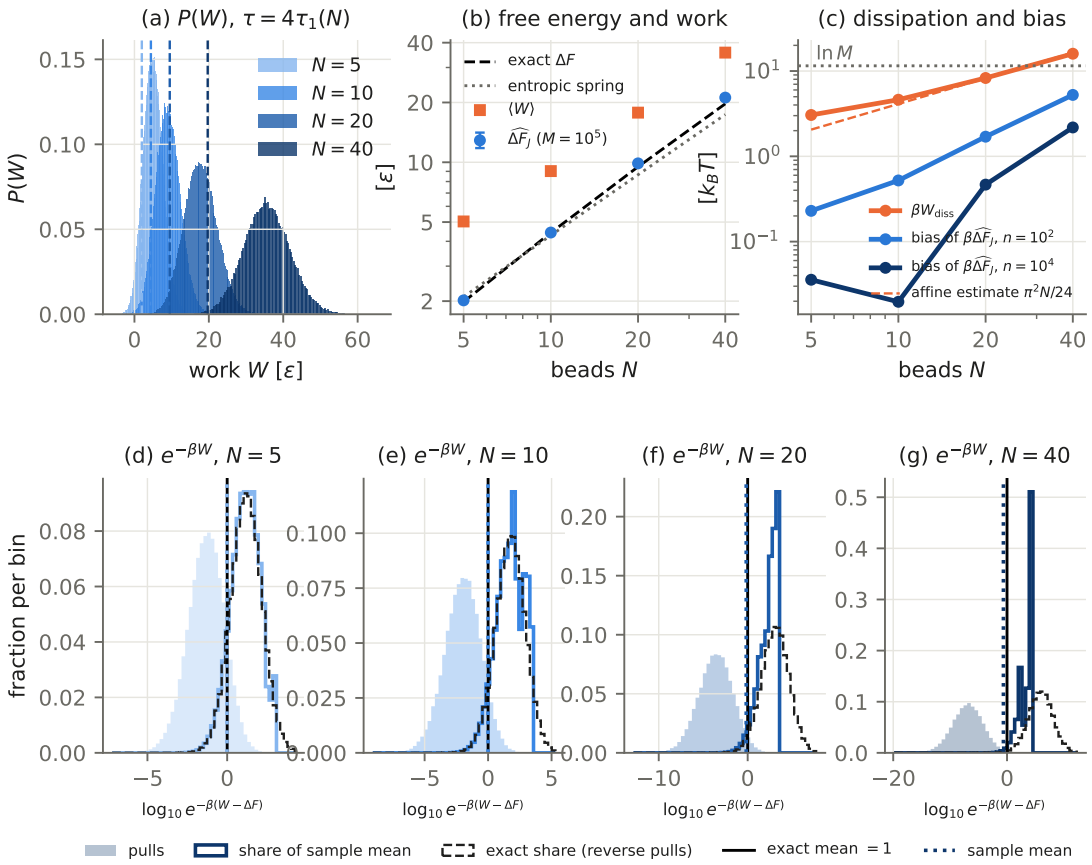


Figure 4: Chain-length dependence at fixed relative extension and fixed reduced speed $\tau = 4\tau_1(N)$, $k_{\mathrm{B}}T = 1$. (a) Work distributions. (b) Exact ΔF, entropic-spring approximation, mean work and Jarzynski estimate. (c) Dissipated work and bias of the Jarzynski estimate with n pulls. (d–g) Distribution of $x = e^{-\beta(W-\Delta F)}$ for each N, with the exact share from reverse pulls.

Third, two-sided estimation changes the scaling of the problem. One-sided estimators need to sample the typical region of the opposite process. BAR needs only overlap between the two, which persists much longer as dissipation grows. In single-molecule experiments, where reverse (refolding) pulls are usually available, this argues for collecting both directions whenever possible, as is done in practice [4].

The model has deliberate limitations. The chain is a phantom chain without excluded volume or bending stiffness, hydrodynamic interactions are neglected, and the motion is two-dimensional. These choices make the exact reference possible. Adding excluded volume would require thermodynamic integration for ΔF, but should not change the qualitative picture, since the scaling argument relies only on the affine dissipation and the extensivity of W_{diss}. The interpretation of the forward–reverse asymmetry in terms of finite extensibility was not tested directly. Varying the bond stiffness or the extension range would test it.

Two natural extensions are protocol optimisation and feedback. Shaping $\boldsymbol{\lambda}(t)$, rather than only its duration, reduces dissipation; in linear response the optimal protocol moves at constant thermodynamic speed. No open-loop protocol can bring the mean work below ΔF, but feedback protocols that react to the measured chain can, at an information cost bounded by the Sagawa–Ueda relation.

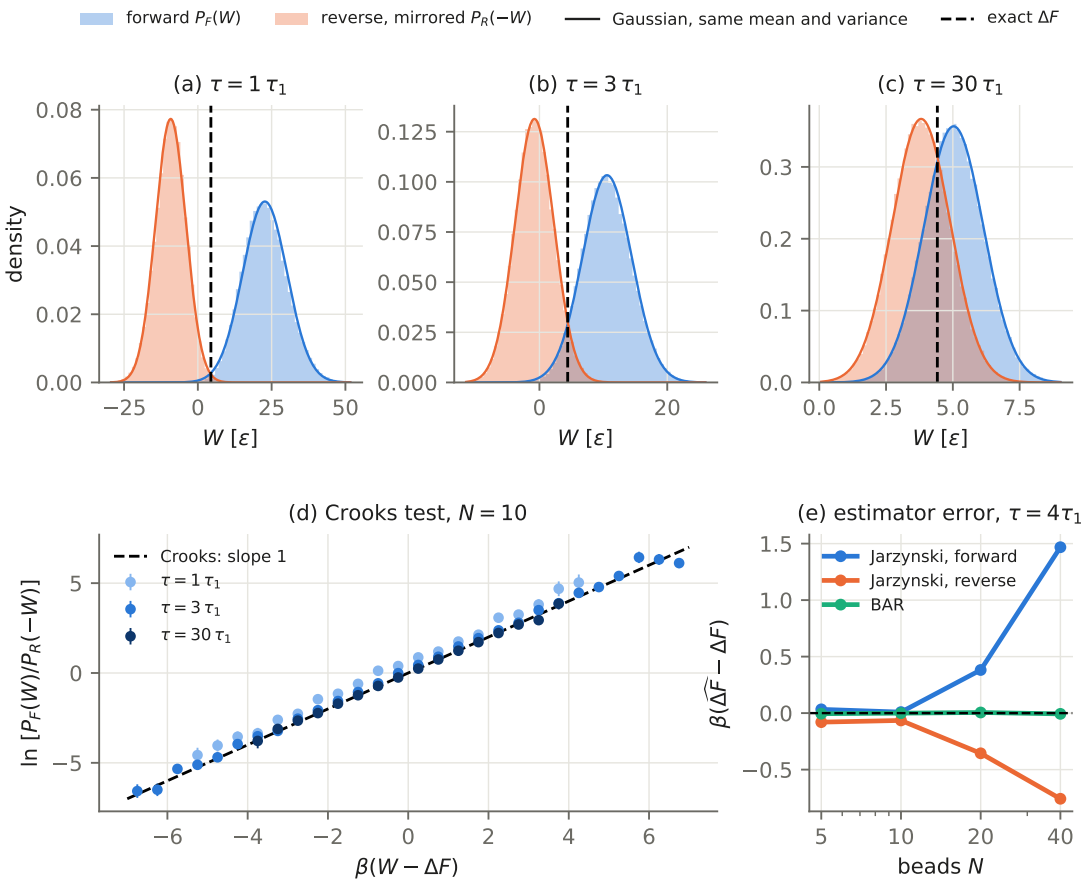


Figure 5: Crooks theorem ($k_\mathrm{B}T = 1$, 10^5 forward and 10^5 reverse pulls per condition). (a–c) Forward and mirrored reverse work distributions for $N = 10$ at three speeds, with Gaussians of equal mean and variance; dashed: $\Delta F = 4.420$. (d) Logarithm of their ratio against $\beta(W - \Delta F)$; the dashed line is the prediction. (e) Error of the forward and reverse Jarzynski estimators and of BAR against chain length.

5 Conclusion

We studied the Jarzynski and Crooks relations for an anchored two-dimensional chain pulled through a Langevin bath, comparing every estimate with exact free energies obtained from a log-space convolution and exact equilibrium sampling. The fluctuation theorems hold to within the statistical and discretisation accuracy of the simulations. Their practical value is governed by the dissipated work, which an affine-stretching argument predicts as $\beta W_\mathrm{diss} \simeq (\pi^2/6)N/(\tau/\tau_1)$. This explains the collapse of work distributions across temperatures at fixed reduced speed and the exponential growth with chain length of the number of pulls the Jarzynski estimator requires. Plots of $e^{-\beta W}$ together with the Crooks identity reveal when the forward estimate is missing most of its average, and Bennett's two-sided estimator recovers the exact free energy where one-sided estimates fail by more than $k_\mathrm{B}T$.

A Numerical checks

Time step. For the Gaussian chain, the exact propagation of the discrete algorithm gives a Jarzynski residual $\langle W\rangle - \beta\sigma_W^2/2 - \Delta F = -8.0\times10^{-4}$, -3.0×10^{-3} and $-1.1\times10^{-2}\,k_\mathrm{B}T$ at $\Delta t = 0.01$, 0.02, 0.04, a clean Δt^2 law. The residual is non-zero because the momentum distribution of BAOAB is slightly distorted. For the stiff chain ($N = 10$, $\tau = 4\tau_1$) halving Δt changed the three-seed mean of $\beta\widehat{\Delta F}_J$ by -0.07 ($k_\mathrm{B}T = 2$) and -0.05 ($k_\mathrm{B}T = 1$), comparable

to the seed-to-seed scatter, and $\langle W \rangle$ by less than 0.1%.

Error bars. For Gaussian work the sampling distribution of $\widehat{\Delta F}_J$ can be simulated directly. At $\beta W_{\mathrm{diss}} = 6.8$ and $M = 10^5$ it has mean bias $+0.10\,k_{\mathrm{B}}T$ and spread $0.28\,k_{\mathrm{B}}T$, three times the bootstrap estimate from a single sample. Independent repetitions of the stiff-chain run at $k_{\mathrm{B}}T = 2$ scatter by $0.09\,k_{\mathrm{B}}T$, twice the bootstrap error. We therefore recommend independent repetitions or block-convergence curves rather than bootstrap errors whenever M_{eff} is below a few thousand.

Reference free energies. The log-space convolution agrees with the closed form for $b = 0$ (64.796 against 64.800 at $N = 40$) and with the Fourier-space formula wherever the latter is numerically reliable. The mean trap force sampled from the exact equilibrium ensemble agrees with $\partial F/\partial\lambda$ from the reference.

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[11] Companion theory document `theory.pdf` (model, three derivations of the Jarzynski equality, Crooks theorem, estimator derivations, numerical scheme), in the same repository.